Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: cpu socket

Assistant: To better understand your requirements for the CPU socket application:

1. What is the exact use of the polymer in the CPU socket? For example, is it for the main rigid housing, contact retention features, or any flexible parts like retention clips?
2. Where will this CPU socket be used? For instance, is it for desktop computers, servers, embedded systems, or another type of device?

Once you provide this information, I will ask about wall thickness, preferred base materials, and openness to alternative suggestions to ensure the best material fit.

User: housing in pc

Assistant: Thank you for clarifying that the polymer is for the main housing of the CPU socket in a PC.

To proceed, could you please provide the following details:

1. What is the minimum wall thickness of the CPU socket housing?
2. Do you have any preferred base materials for this application (for example, specific polymers you have used before)?
3. Are you open to alternative material suggestions if there is a better fit for your requirements?

User: 0.8 mm thickness other information i dont have i am open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your CPU socket housing application:

1. Minimum wall thickness: 0.8 mm (as specified)
2. Flame rating: UL 94 V-0 at 0.8 mm
3. Processing: Injection molding
4. Dimensional stability
5. High flow (for thin-wall molding)
6. Warpage resistance / housing flatness
7. Lead-free soldering resistance
8. Low dissipation factor (Df) for electrical performance
9. Dielectric property alignment (for signal integrity)

Are all of these requirements correct for your application? Please confirm or let me know if any need adjustment.

User: open to suggestions

Assistant: To further refine the material selection, could you please confirm or provide your requirements for the following:

1. Relative Thermal Index - Electrical (RTI Elec): Is there a minimum temperature rating you require for long-term electrical performance?
2. Comparative Tracking Index (CTI): Do you have a minimum CTI value needed (for example, 250 V, 400 V, or 600 V)?
3. Glow Wire Ignition Temperature (GWIT) and Glow Wire Flammability Index (GWFI): Are these required for your application, and if so, at what values?

If you are unsure or have no specific requirements for any of these, please let me know.

User: i dont ave this information recommedn the grades based on what we discussed

Assistant: Based on your requirements for a CPU socket housing in a PC (0.8 mm wall thickness, UL 94 V-0, high flow, dimensional stability, and lead-free soldering resistance), here are some potential options that may be suitable for your application:

1. ZENITE® 5145L – Excellent fit for CPU socket housings, UL 94 V-0 at 0.8 mm, high flow, low warpage, and lead-free soldering resistance.
2. ZENITE® 5130L – Also meets UL 94 V-0 at 0.8 mm, high flow, improved toughness, and good dimensional stability.
3. VECTRA® E130i – Very high flow for thin-wall molding, UL 94 V-0 at 0.2 mm and above, and proven reflow performance.
4. ZENITE® 5115L – Meets UL 94 V-0 at 0.79 mm, high flow, and favorable dielectric properties.
5. VECTRA® S135 – High heat resistance, very low moisture absorption, and excellent dimensional control.
6. VECTRA® S475 – Super high flow and low warpage, suitable for very thin or complex features.
7. ZENITE® 251 – Very low dielectric constant and dissipation factor, ideal if signal integrity is a top priority.

These grades are selected based on their technical properties and historical use in similar applications. Please note that final material selection and product design decisions should be validated with your engineering team and through prototyping. Ensure compliance with all applicable standards and regulations for your end-use application. If you need more detailed technical data or have specific regulatory requirements, I recommend connecting with a live Celanese materials specialist for further support.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.